

NEMSIS: Data In and Data Out

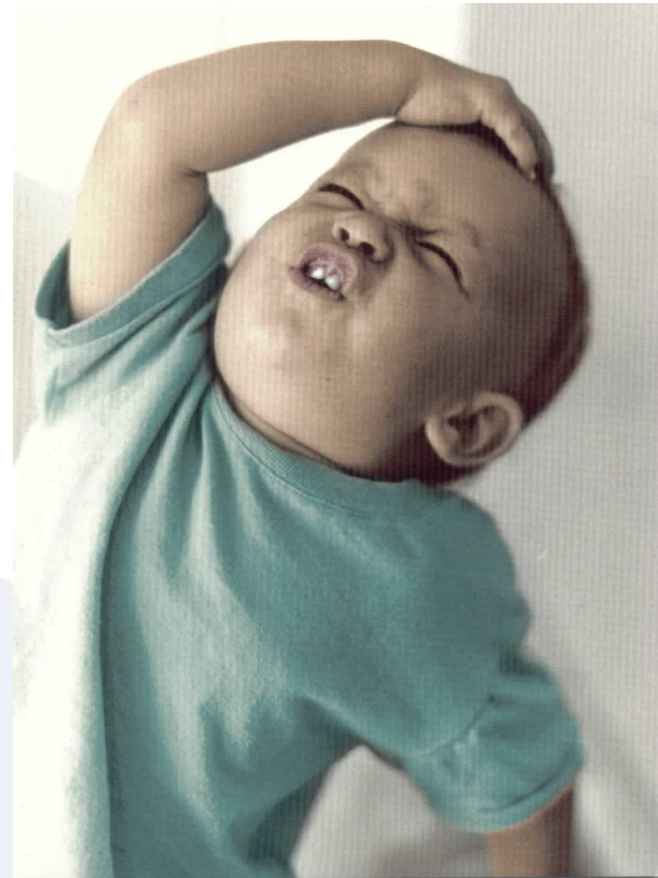
Evaluating and Improving Performance Through
EMS Data

David Owens

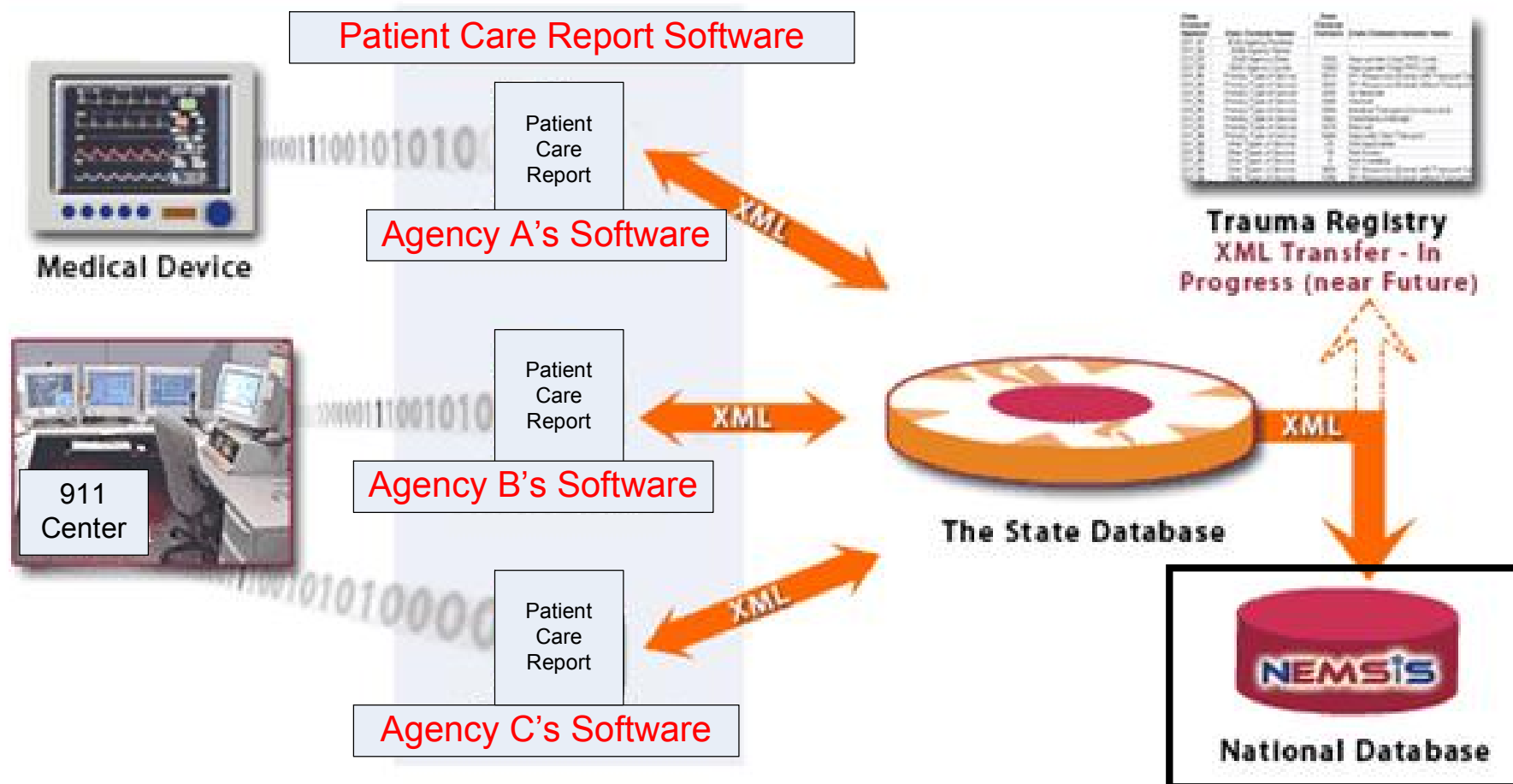
Director of the NEMSIS Technical Assistance Center



Where are we?



THE PORTABILITY OF DATA - the **NEMSIS** STANDARD.

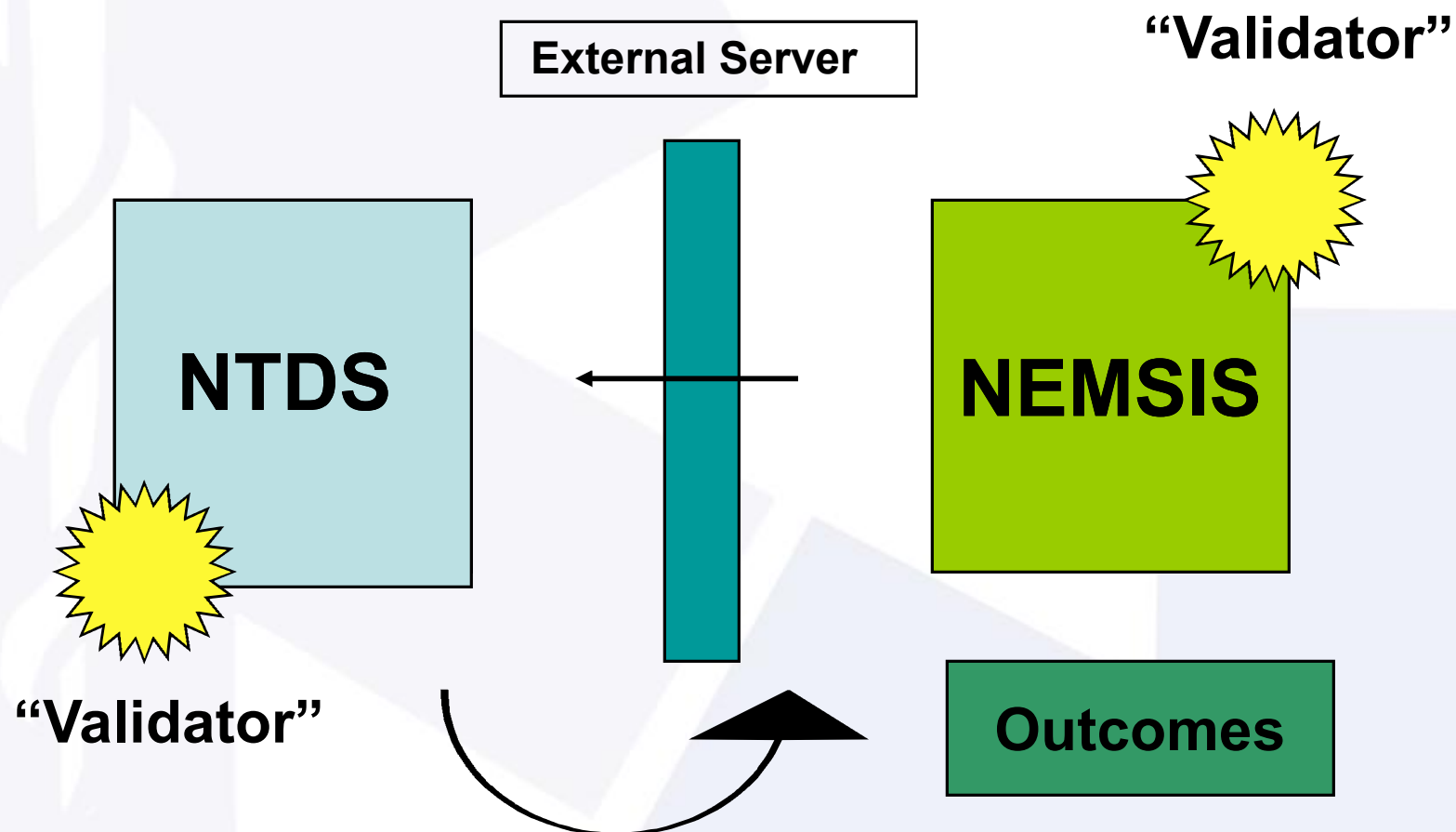


National Trauma Data Standard



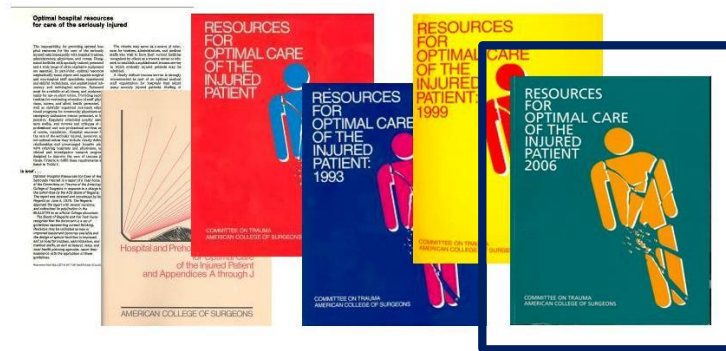
Revised May 2007

Benefit of a Common XML



Verification Requirement

- Annual NTDB participation
- Officially in effect with new green book



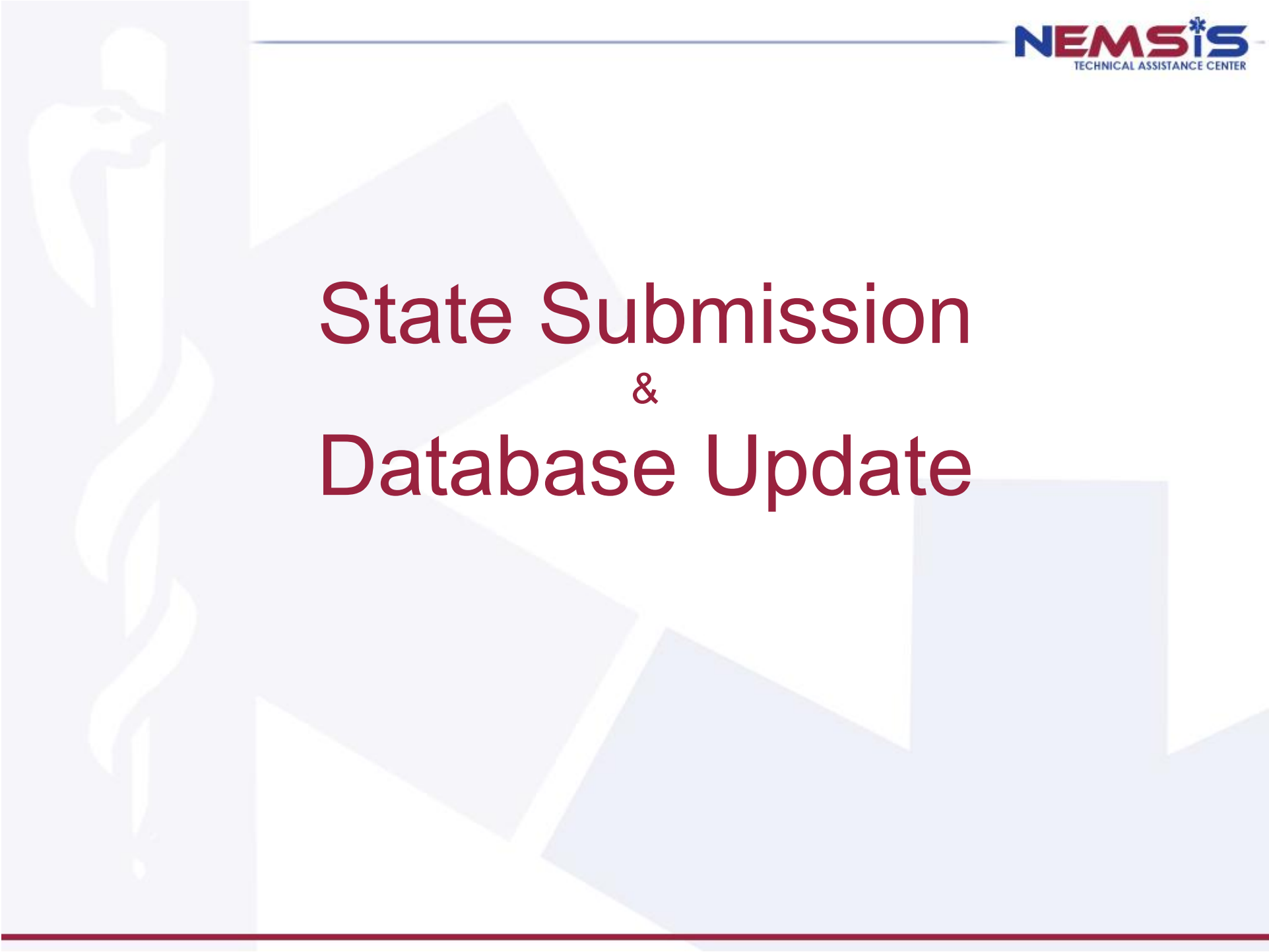
- The first call for NTDS data (2007).

Technical Assistance to States

- Direct Technical Assistance
 - Conducted site visits in 21 states
 - 2 additional state contacts while conducting another visit
 - 48 states received direct assistance
 - Participation in state meetings
 - Creation of state-specific tools (i.e., Validator and Mapper software tools)
 - Collaboration with NEDARC

Technical Assistance to States

- Indirect Technical Assistance
 - NEMSIS website accessed over 48,000 times
 - 107,000 page views
 - Software Contact List
 - Wolfberg Letter (HIPAA)
 - National Reports
 - List Server Messages / Google User Group
 - Software Compliance Testing

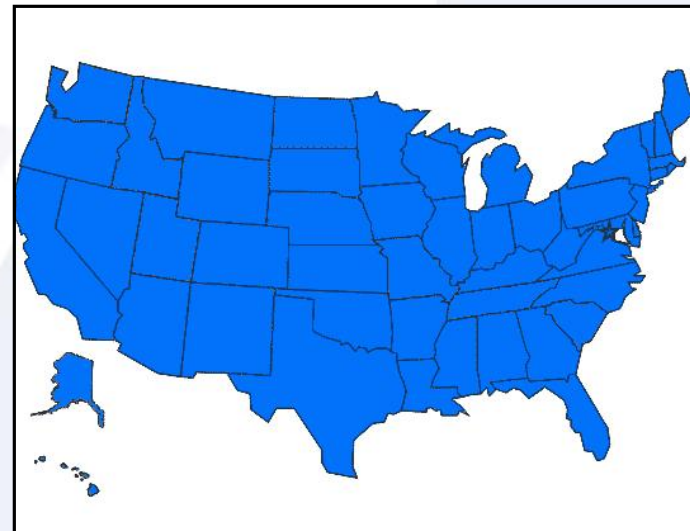


State Submission & Database Update

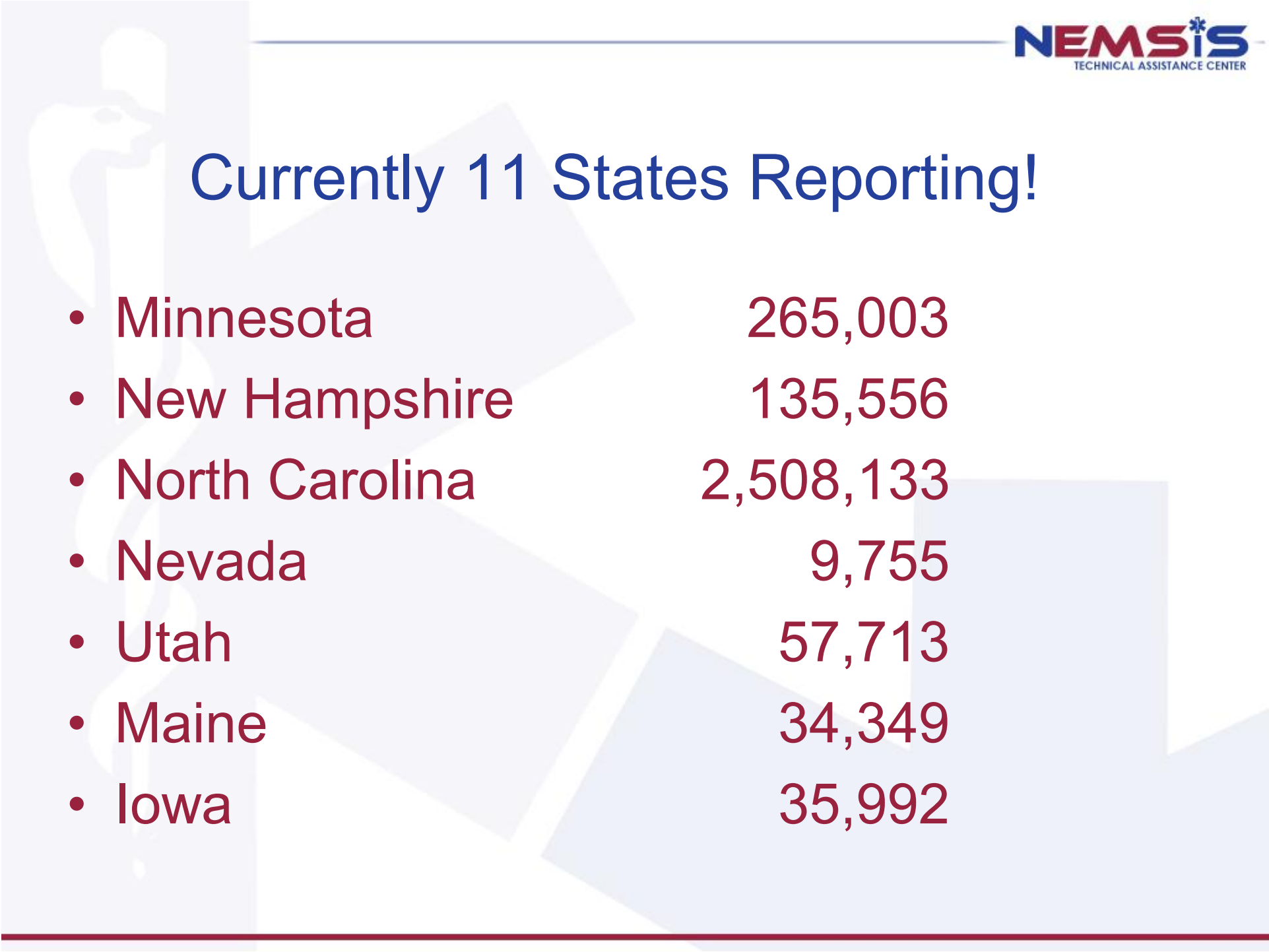
Financial Assistance to States

- States Requesting 408 Funding for NEMSIS
 - AL, AR, **AS**, AZ, CA, CO, CT, FL, HI, ID, IN, KS, LA, ME, MI, MN, MO, MS, MT, NE, NJ, NM, OH, OK, PA, **PR**, RI, SD, TN, UT, VT, WA, WI, WY.

- 32 States
- 2 Territory



Currently 11 States Reporting!



• Minnesota	265,003
• New Hampshire	135,556
• North Carolina	2,508,133
• Nevada	9,755
• Utah	57,713
• Maine	34,349
• Iowa	35,992

Update Con't

• North Dakota	18,323
• Nebraska	28,751
• Florida	<u>110,564</u>
Total	3,204,139
• Oklahoma (test files only)	1

State	2006				2007				2008				Total Rows
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1				by State
Florida							9,496	53,009	48,059				110,564
Iowa					18,006	17,986							35,992
Maine					8,175	8,216	10,780	7,178					34,349
Minnesota	26,816	29,340	19,436	29,108	38,289	47,477	46,274	28,263					265,003
Nebraska					6,525	6,543	9,002	6,681					28,751
Nevada					5,835	3,920							9,755
N Hampshire	16,282	18,706	21,837	20,411	19,669	18,806	19,845						135,556
North Carolina	507,801	350,558	380,182	358,280	195,259	223,333	245,200	247,520					2,508,133
North Dakota					9,100	9,223							18,323
Utah					10,530	12,438	14,588	11,635	8,522				57,713
Total Records													3,204,139

States Poised to Submit in First 6 Months of 2008

- Idaho
- Hawaii
- Missouri
- Georgia

Remainder of States in 2008

Potential Total of 20 by Year End

- Delaware
- West Virginia
- Marianas Islands
- Wisconsin
- Guam
- Michigan
- Michigan
- Alabama
- Montana
- Washington
- Tennessee
- New Mexico



Overview of the NEMSIS Reporting System

National Reporting Plan

- **Develop “real time” national reports**
 - **Report Domain**
 - Data Quality reports
 - Demographic reports
 - Elapsed Time reports
 - Medical and Trauma reports
 - Disposition reports
 - Agency reports
 - Cardiac Arrest reports
 - Financial Aspects reports
 - Delay reports
 - Ad-hoc reports
- **Develop a “real-time” searchable data query system for emphasis areas**
 - Cardiac Arrest
 - Trauma
- **Develop “real-time” data cubes to facilitate multi-dimensional data models with a rapid execution time.**

State-level Reporting Plan

- **Develop “real time” state reports with national benchmark**
 - **Report Domain**
 - Data Quality reports
 - Demographic reports
 - Elapsed Time reports
 - Medical and Trauma reports
 - Disposition reports
 - Agency reports
 - Cardiac Arrest reports
 - Financial Aspects reports
 - Delay reports
 - Ad-hoc reports
- **Develop a “peer benchmark group” for additional comparisons**
 - Three to five states chosen by state leadership
- **Develop “real-time” data cubes to facilitate multi-dimensional data models with a rapid execution time.**

Agency Reporting Plan

- **Provide “real time” report schemes to Vendors**
 - **Report Domain**
 - Data Quality reports
 - Demographic reports
 - Elapsed Time reports
 - Medical and Trauma reports
 - Disposition reports
 - Agency reports
 - Cardiac Arrest reports
 - Financial Aspects reports
 - Delay reports
 - Ad-hoc reports
- **Develop “agency profiles” posted on the web for benchmarking**
 - Profiles based upon NEMSIS and US Census data

Overview of the
NEMSIS Data
Revision Plan

Revision to NHTSA 3.0

- The revision process will begin Spring 2008
 - Lead by Greg Mears
 - Fix existing errors in the 2.2.1 dataset
 - Consider the additional of modules (e.g., workforce)
 - Facilitate national research (e.g., CDC triage scheme)
 - Process will be more electronic
 - Wiki, teleconferences, involve vendors
 - Product available in 2010

NEMSIS Data Dictionary v3 Wiki



NEMSIS
TECHNICAL ASSISTANCE CENTER

navigation

- [Main Page](#)
- [Data Dictionary](#)
- [Add General Comment](#)
- [Add Variable Specific Comment](#)

Main Page

Welcome to the NEMSIS Data Dictionary Revision Wiki! The purpose of this Wiki is to provide a way for folks to comment on changes or additions to the NEMSIS Data Dictionary. This wiki allows you to provide general comments regarding the structure or content of the NEMSIS data set. You may also provide comments specific to a particular variable currently in the NHTSA 2.2.1 data set. Comments provided on this Wiki will be used to revise the NHTSA 2.2.1 data set to the NHTSA 3.0 data set. This wiki will accept comments until April 15th, 2008. A schedule associated with the formal NHTSA revision process will be provided in the near future.

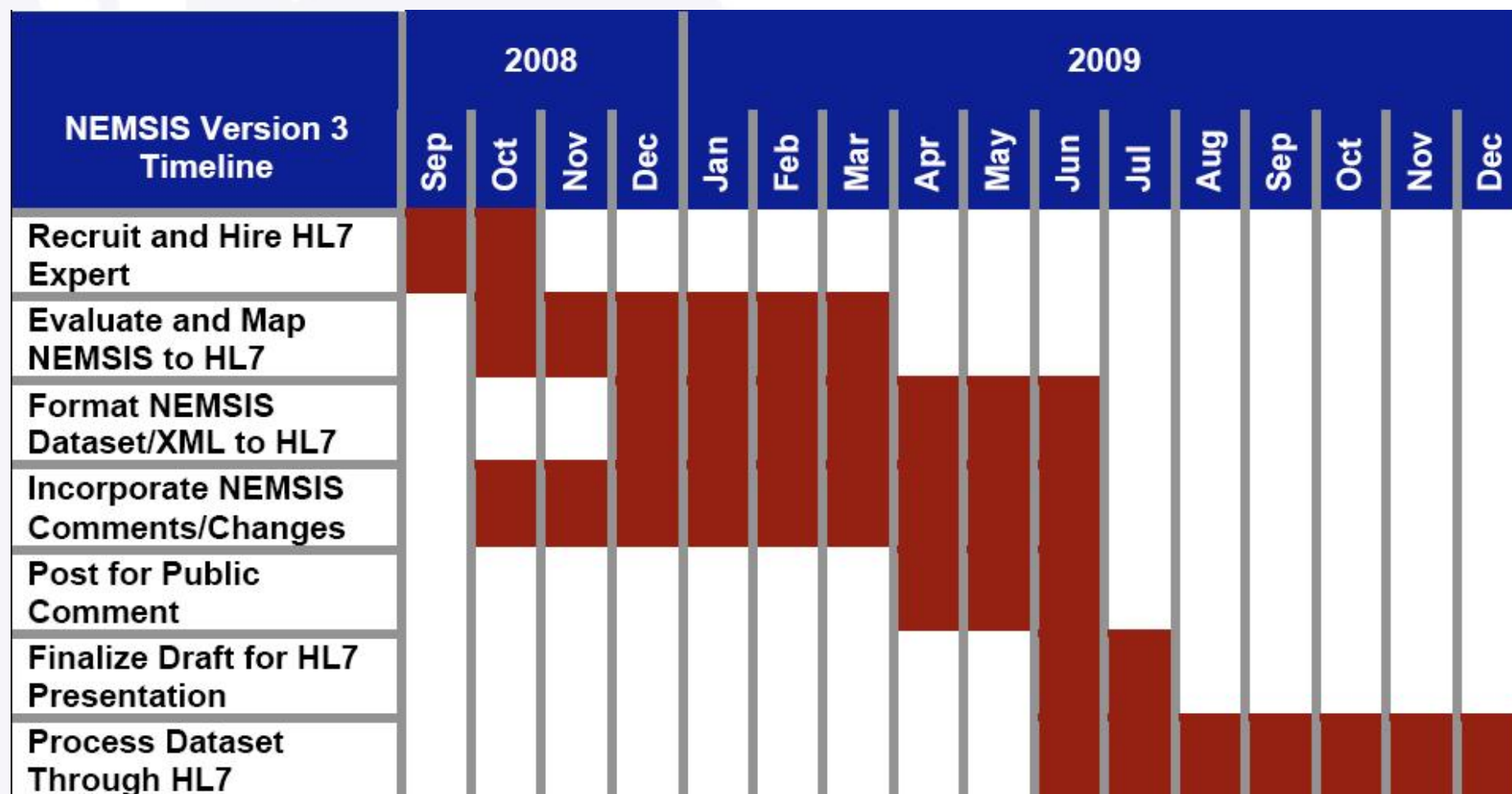
Click [Data Dictionary](#) to begin contributing.

This page was last modified 22:48, 4 January 2008. This page has been accessed 225 times. [Privacy policy](#)

[About NEMSIS TAC Data Dictionary v3 Revision](#) [Disclaimers](#)



NEMSiS Version 3 Timeline



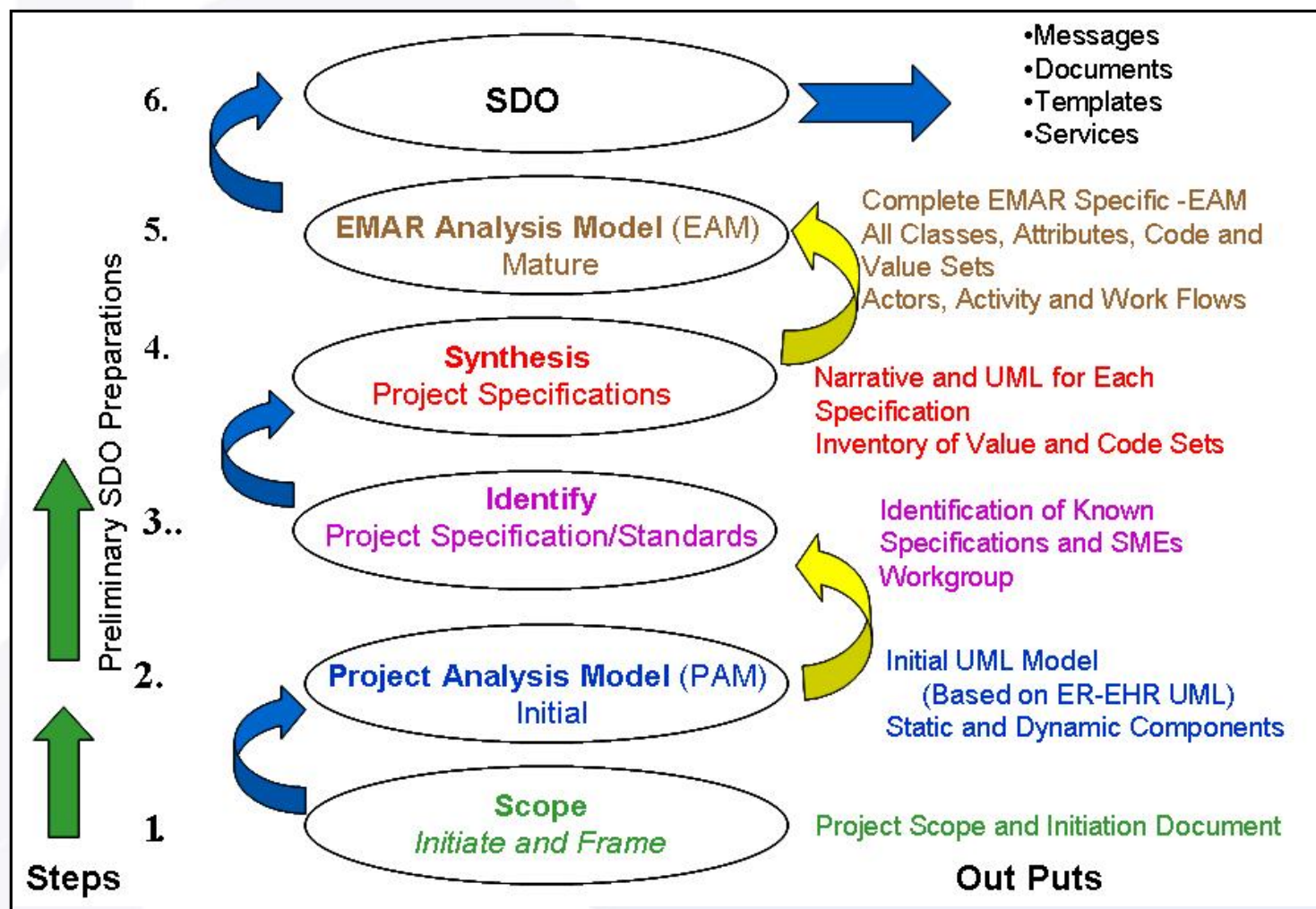


Update regarding SDO Process

Development of the Gap Closure Process

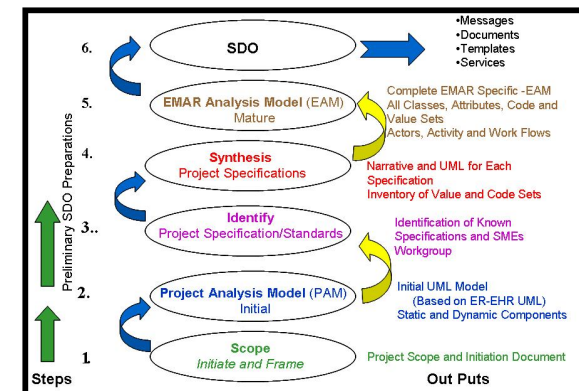
- Participated with the ER-EHR Use Case
 - Asked NEMSiS to develop gap closure process
 - Developed of the HITSP ER-HER (IS04) Gaps and Overlaps Closure Plan
 - Organizes the Emergency Medical Awareness and Response (EMAR) Leadership Group
 - Greg will serve as Chair
 - Defines process for Gap Closure

EMAR Process



NEMSiS SDO Plan

- We are PROJECT #6!
- Will use the EMAR process



- Meeting in Dec. (SDO and HL7 experts)
- Developed draft plan and budget (HL7)
- Reviewed by NHTSA, funds targeted!
- Project scope approved by HL7 (Patient Care)

Tasks Seem Overwhelming?



Technical Assistance



- Visit the websites
 - Join the Google Groups
- Talk with your software vendors
- Utilize resources at NEMSIS TAC and ACS



SEARCH NEMSIS:

Download Da

Home

The Project

The Partners

Technical Support

Data Elements

Compliance

Reference Materials

Contact Info



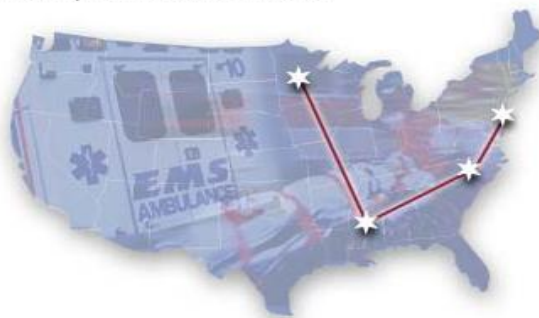
The NEMSIS Technical Assistance Center (TAC)...

Offering technical assistance on the implementation and tracking of National EMS Data.

"We are much stronger as one, then we will ever be as many."

-Ed Racht, MD
Austin-Travis County EMS

(Speaking in reference to the heroic response made by the Texas EMS system during the hurricanes of 2005.)



www.NEMSIS.org

Latest News

Download the Latest NEMSIS IHTSA Dataset Version 2.2.1

Need the latest dataset dictionaries and schemas? Find it all here.

The Third Round of Compliance Testing has Been Completed!

There are now 18 software developers that are either Silver or Gold compliant

The NEMSIS TAC Releases a Minor Version Change for the NEMSIS IHTSA Dataset

Due to the discovery of a number of issues from those using the dataset, we have made a minor version change. Please click on the above link to read more.

NEMSIS System Animation

A visual animation created to explain the NEMSIS system.

Software Vendors

Software Developer's Meeting Presentations

Click here to obtain copies of the presentations made at the meeting.

Compliance Information

Click here for important news about the compliance testing process.

Software Compliance Testing and Certification

Click on the above link to learn how to certify your EMS software package as being NEMSIS Compliant.

NEMSIS Validator Application

Download the Validator Software, test files, and a quick reference guide.

Available Resources



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National Report Sample

National Elapsed Patient Care Times By Agency Type*

A National EMS Database Report

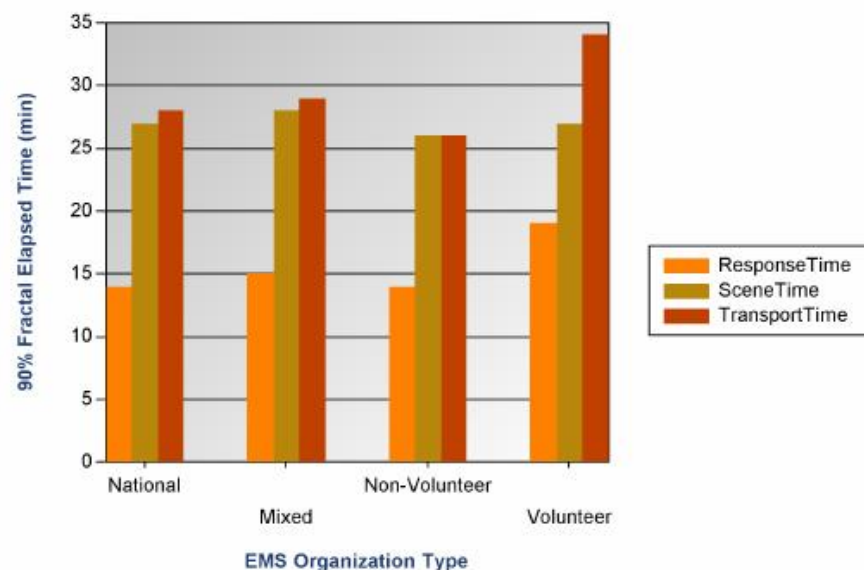
States Submitting Data: Minnesota, New Hampshire and North Carolina †

Date Range: 01/01/2006 - 12/31/2006

Record Selection: 911 Response Time (Unit Notified by Dispatch Time until Unit Arrives at Scene Time) using Lights and Sirens

Records Used: 300,680 ‡

90% Fractal Elapsed Time By Type of EMS Organization



National Report Sample

The following table provides added statistical descriptions of the information used to create the chart labeled: National Elapsed EMS Patient Care Times By Agency Type.

EMS Organization	Total Records	Patient Care Times	Average	Median	90% Fractal
National					
	300680	ResponseTime	00:09:59	00:07:00	00:14:00
	261390	SceneTime	00:17:10	00:15:00	00:27:00
	234776	TransportTime	00:15:11	00:12:00	00:28:00
Mixed					
	38121	ResponseTime	00:08:33	00:06:00	00:15:00
	36452	SceneTime	00:17:56	00:16:00	00:28:00
	31036	TransportTime	00:15:20	00:12:00	00:29:00
Non-Volunteer					
	236464	ResponseTime	00:10:03	00:07:00	00:14:00
	199611	SceneTime	00:17:12	00:15:00	00:26:00
	181178	TransportTime	00:14:39	00:12:00	00:26:00
Volunteer					
	26095	ResponseTime	00:11:30	00:09:00	00:19:00
	25327	SceneTime	00:15:55	00:13:00	00:27:00
	22562	TransportTime	00:19:14	00:16:00	00:34:00

Footnotes:

* Refer to the linked document for detailed specifications associated with this chart:

[National Elapsed Patient Care Times report specifications](#)

† States vary in the inclusion criteria used to populate EMS Data Sets. Please see the NEMSIS website for a description of differences in state EMS database composition.

<http://www.nemsis.org/nemsisReporting/nationalEMSDatabaseDescription.html>

‡ For the purposes of charting, all elapsed patient care times that were negative, zero, or greater than 24 hours were removed. Overall, the following records were excluded from the analysis:

6% of the Response Time records,
20% of the Scene Time records and
28% of the Transport Time records

National Report Sample

National Elapsed Patient Care Times by Agency Type National EMS Database Report Specification

Purpose

This report details national information regarding the EMS elapsed time to patient care for several common response intervals by type of EMS organization.

Definitions

1. **Type of EMS Organization:** Common EMS organizational types are divided into 3 groups based on reimbursement for EMS treatment rendered. The terms used are general in nature and are for National EMS Database Reports only. Terms do not reflect state or agency terminology. For more descriptive definitions, contact individual participating states or EMS agencies. The groups are divided as follows:

Volunteer: A volunteer agency employs properly trained individuals and operates an emergency response vehicle to or from the scene of an emergency and/or renders emergency medical treatment on a volunteer basis.
Non-Volunteer: A non-volunteer agency employs properly trained individuals and operates an emergency response vehicle to or from the scene of an emergency and/or renders emergency medical treatment and seeks reimbursement for services offered.
Mixed: A mixed agency includes a core of compensated employees who are supported by true volunteer personnel.
2. **Average Value:** The average is the sum (total amount) of all of the numbers divided by the number of items for a specific measurement.
3. **Median Value:** The number in the middle of the distribution of numbers. It is the point such that half of the numbers fall below and half above.
4. **90% Fractal Value:** The value or measurement at which 90% of all events occur. This is typically used in time measurements to better standardize performance across systems.
5. **Lights and Sirens:** A lights and sirens, emergent response to or from an EMS event.
6. **No Lights and Sirens:** A normal traffic speed response (no lights and sirens) to or from an EMS event.

National Report Sample

Required Data Elements

- * E02_04: Type of Service Requested
- * E02_20: Response Mode to Scene
- * E05_04: Unit Notified by Dispatch Date/Time
- * E05_06: Unit Arrived on Scene Date/Time
- * E05_09: Unit Left Scene Date/Time
- * E05_10: Patient Arrived at Destination Date/Time

1. The Unit Notified Date (E05_04) is used to select the records for the Date Range
2. Select only records with E02_04 = 911 Response (Scene)
3. Select only records with E02_20 = Lights and Sirens

Formulas

Response Time = E05_06 – E05_04

Scene Time = E05_09 – E05_06

Transport Time = E05_10 – E05_09

Exclusions/Limitations

For the purposes of charting, all elapsed patient care times that were negative, zero, or greater than 24 hours were removed.

National Report Sample

EMS INCLUSION CRITERIA FOR STATES	North Carolina	New Hampshire	Minnesota	Nevada
(x = data are available in state EMS registry)				
Any 911 Call				
with an EMS dispatch		X	X	X
without an EMS dispatch				X
Any 911 EMS Dispatch				
with patient contact	X	X	X	X
without patient contact			X	X
Any EMS Response				
with patient contact	X	X	X	X
without patient contact			X	X
Any EMS Patient Contact				
with evaluation/care	X	X	X	X
without evaluation/care	X	X	X	X
with transport	X	X	X	X
without transport	X	X	X	X
Any ALS Care				
with transport	X	X	X	X
without transport	X	X	X	X
any EMS transport	X	X	X	X
Any Medical Transports				
consider non-emergent	X		X	X
scheduled transports	X		X	X
Any Ground-Based Inter-Facility Transports				
ground transports	X	X	X	X
Air (rotorwing) transports	X	X	X	X
any long-distance air medical transports (fixed wing)	X		X	X
What other EMS- Related Events are Included in the State Data System?				
no other events				
first responder reports	X	X		
rescue events	X	X		
standby requests	X	X	X	X
mutual aid	X	X	X	X
How often is EMS Record Data Provided to the State?				
	Daily	Daily	Daily	Daily
Is the Data Provided to the State ever Changed Later?				
	Yes	Yes	Yes	Yes
In your Opinion, What Percent of the EMS Agencies in the State are Providing Data?				
	100%	81%	100%	80%
In your Opinion are the EMS Agencies which Provide Data to the State Providing 100% of their EMS Events?				
	Yes	Yes	Yes	Yes
If No, in your Opinion, what Percentage of all EMS Events meeting State Criteria are Actually Submitted to the State?				